

Module 14

Accounting for Divisions

Learning Objectives

By the end of this module you should understand:

1. The reasons for divisionalization
2. The advantages and shortcomings of divisionalization
3. The types of divisional structures
4. The role of return on investment and the difficulties in measuring both return and investment
5. The weaknesses of ROI:
 - how residual income is calculated and the differences between RI and ROI;
 - how well-managed companies have a wide variety of performance measures;
 - the role of transfer pricing in divisionalized companies and the difficulties inherent in selecting suitable bases.

Agenda

14.1 Introduction

14.2 Why Divisionalize?

14.3 Types of Divisions

14.4 Defining Profits and Investments

14.5 Asset Base Valuation

14.6 Residual Income: An Alternative to ROI

14.7 The Imputed Rate of Interest Does Matter!

14.8 A Cautionary Note about Performance Measures

14.9 Transfer Pricing

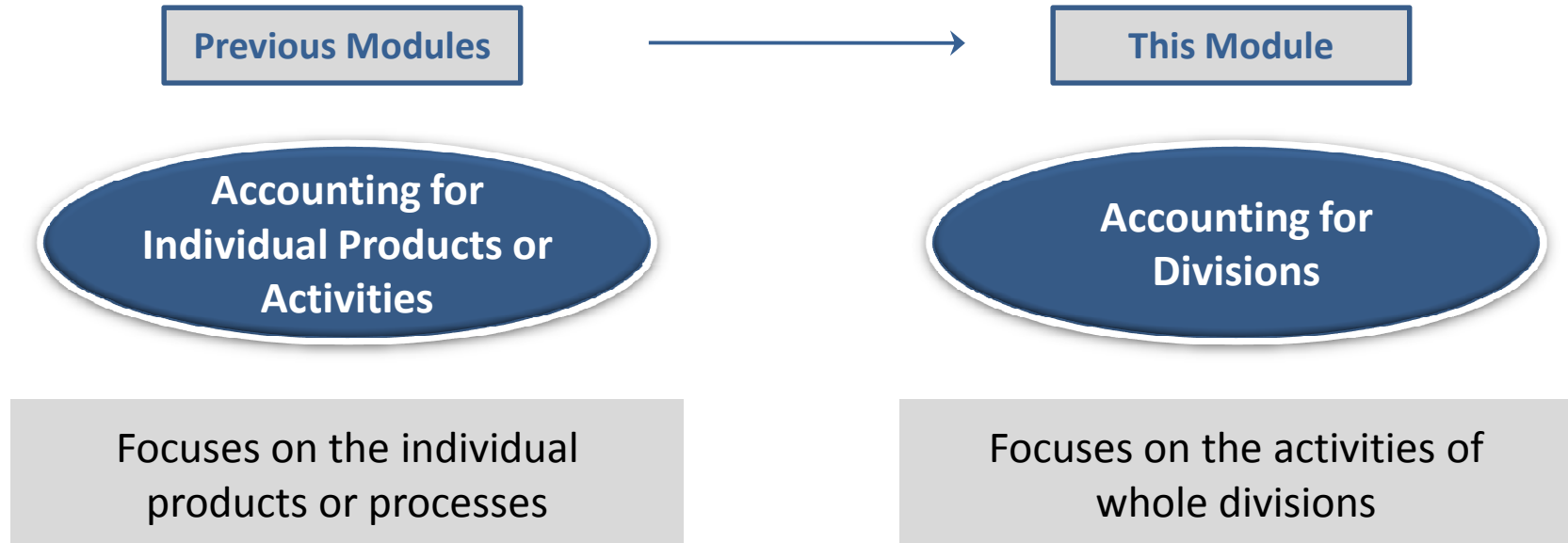
14.10 Criteria for Establishing a Transfer Price

14.11 The International Dimension

Review Questions

Case Studies

In this module, we'll study costing for companies whose activities are sufficiently diverse to the extent that it leads to separating divisions



We will turn our attention to the accounting issues surrounding internal transfers of goods and services in organizations

Take MBA as an example of a divisionalized company

From its 20x2 Annual Report (see (Appendix 5.1) we can see that its operations are structured around two principal sectors:

1. Industrial operations
2. Financial operations

A wide spectrum of research and manufacturing activity is conducted all over the world and reported in the Annual Report (not reproduced in Appendix 5.1) in four major blocks, United Kingdom, Rest of Europe, The Americas and Africa, Asia and Australasia. Our interest lies in the problems which arise when one division of a company sells products to another division of the same company instead of to outside customers.



Do the same prices apply? Can the selling division be allowed to make a profit from the transaction? What would happen if the buying division could source the product from an external supplier more cheaply? Additionally we will consider how to evaluate the overall performance of individual divisions. Can we use the same procedures for evaluating divisions as we use for entire companies?

A division is sometimes a wholly owned company; where we also apply the same principles

Agenda

14.1 Introduction

14.2 Why Divisionalize?

14.3 Types of Divisions

14.4 Defining Profits and Investments

14.5 Asset Base Valuation

14.6 Residual Income: An Alternative to ROI

14.7 The Imputed Rate of Interest Does Matter!

14.8 A Cautionary Note about Performance Measures

14.9 Transfer Pricing

14.10 Criteria for Establishing a Transfer Price

14.11 The International Dimension

Review Questions

Case Studies

Businesses grow in two ways, organically or through acquisition

Organic Growth

Driven by more and more sales of more and more products

The increasing cash flow and profits are invested in more productive capacity

Growth is usually slow

The risks attached are relatively small

Due to the detailed knowledge possessed by the company in the areas earmarked for expansion

Growth Through Acquisitions

Acquisitions are financed either by own cash, borrowed cash or issue of own shares

By retaining the management of the enterprise bought, they 'learn the business' and eventually add value to its operations

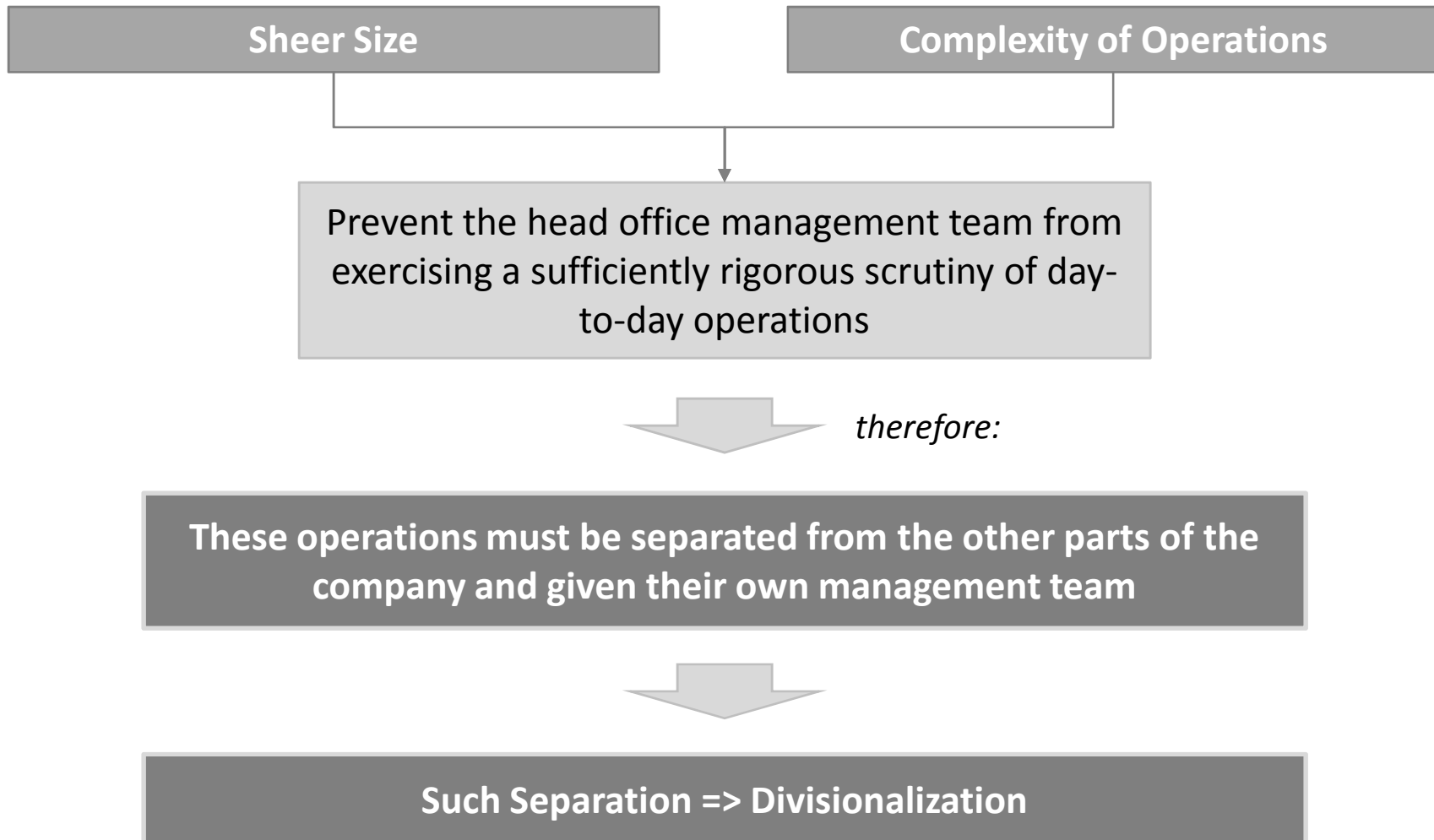
Growth is quick

Growth into areas in which the company is less certain

The two forms of growth are not mutually exclusive

Whichever form of growth a company chooses, it is faced with managerial problems of control

Reasons Behind This:



Divisionalization has many advantages and few disadvantages that are claimed here

Advantages

Specialization

The sheer complexity of MBA's group activities makes the point rather clearly for specialized management in different parts of the company

Size

Even in one-product companies, the magnitude of the problems confronting management may be so great that it makes sense to separate the activities into divisions, either on a geographical basis or a customer basis

Motivation

A feeling of 'ownership' and pride can be engendered for the benefit of the whole company by a divisionalized structure where individual divisions are instructed to treat themselves like companies operating on their own account and be judged according to their own performance

Sharper Decisions

Were problems arise, local solutions can be sought and implemented so much more quickly than if the HQ staff had to review the matter, consult with the local manager, make the decision and communicate the decision to the local team

Career Mobility

Divisions provide a testing ground for managerial talent as it allows the headquarters team to be drawn from the best available managerial talent, a divisional structure is advantageous

Divisionalization has many advantages and few disadvantages that are claimed here

Disadvantages

Lack of Control

Once a division has been set up and its management team given the task of 'running it, headquarters has lost a measure of control



This is overcome by designing a structure for reporting and discussion which allows headquarters to maintain control over the major decisions while encouraging local management to run the routine operations

Cost

Divisionalization is seen to cost more money; as once a division is set up to be, in effect, a company on its own, it will require support services such as personnel, training, computing, perhaps R&D

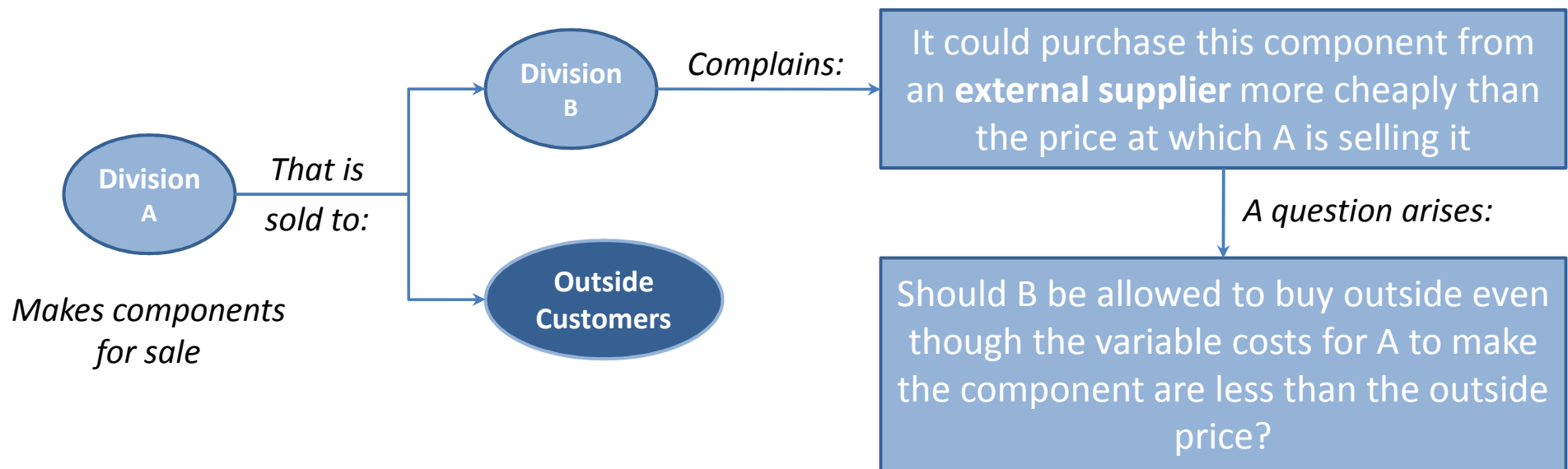


Some companies cut headquarters' overheads on the basis that a significantly reduced set of services will be expected from HQ now that the divisions have their own support

Internal Rivalries

If divisions are set up with a performance measurement system which can highlight superior managerial performance, a spirit of competition and internal rivalry may be created which is dysfunctional to the overall objectives of the company

Example for how internal rivalries take place in a company



From Division B's point of view, the external price is the better one; from the company's point of view the internal cost is the better

Agenda

14.1 Introduction

14.2 Why Divisionalize?

14.3 Types of Divisions

14.4 Defining Profits and Investments

14.5 Asset Base Valuation

14.6 Residual Income: An Alternative to ROI

14.7 The Imputed Rate of Interest Does Matter!

14.8 A Cautionary Note about Performance Measures

14.9 Transfer Pricing

14.10 Criteria for Establishing a Transfer Price

14.11 The International Dimension

Review Questions

Case Studies

Businesses may divisionalize on the basis of sales territories, product lines, production plants, retail outlets, customer layers or any identifiable segment of business

Four main groupings can be identified from the accounting stance:

Cost Centers

Divisions are held responsible only for the incurrence of cost and have nothing to do with the generation of associated revenues



Examples

Government-funded research laboratory

Revenue centers

Divisions are given the task of generating funds without any reference to the underlying costs or disposition of those funds



Sales division of a shipping company

Profit Centers

Costs and revenues are matched and divisional performance is assessed on bottom-line profit in the conventional manner



All divisions are applicable

Investment centers

Divisions' net assets are taken into account when evaluating profit performance



All divisions are applicable

Investment centers allow corporate headquarters to assess not only the profitability of each division but also the efficiency with which assets are employed by the divisions

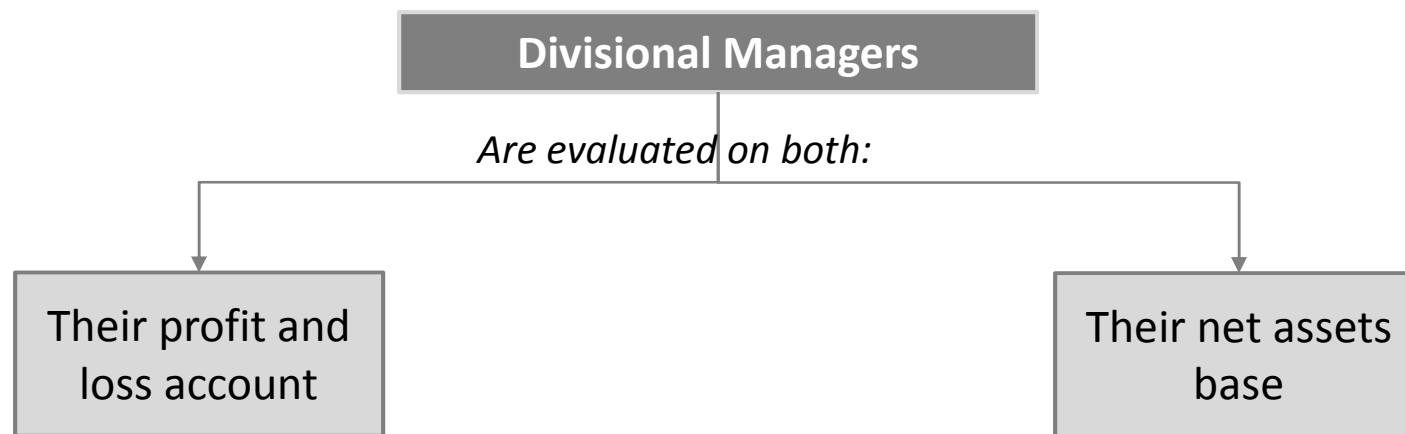
Example:

Division A may report a profit of €10 million and Division B, €8 million

	Division A	Division B
Profit	€10m	€8m
Net assets invested	€100m	€60m
Return on investment	10%	13.3%

The conclusion that A is a more profitable division than B may have to be changed when the investment in net assets is considered alongside profit

Our interest in this module will be concentrated on investment centers



It should not be inferred from the term 'investment centre' that the divisional management has unfettered freedom over investment in assets



Corporate headquarters will have to approve (and perhaps provide the funds for) investment proposals over a predetermined threshold

Agenda

14.1 Introduction

14.2 Why Divisionalize?

14.3 Types of Divisions

14.4 Defining Profits and Investments

14.5 Asset Base Valuation

14.6 Residual Income: An Alternative to ROI

14.7 The Imputed Rate of Interest Does Matter!

14.8 A Cautionary Note about Performance Measures

14.9 Transfer Pricing

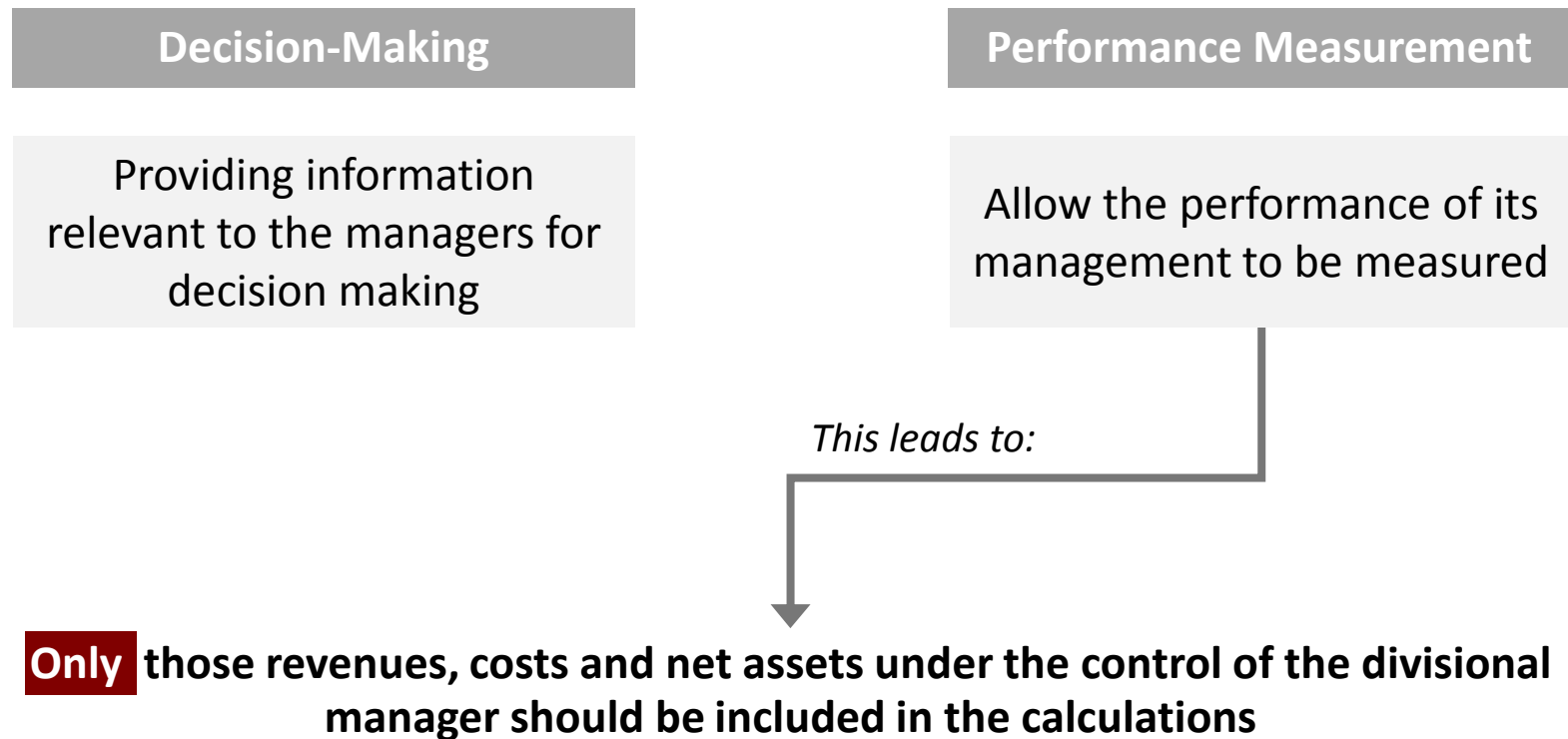
14.10 Criteria for Establishing a Transfer Price

14.11 The International Dimension

Review Questions

Case Studies

There are two aims for divisional accounting systems

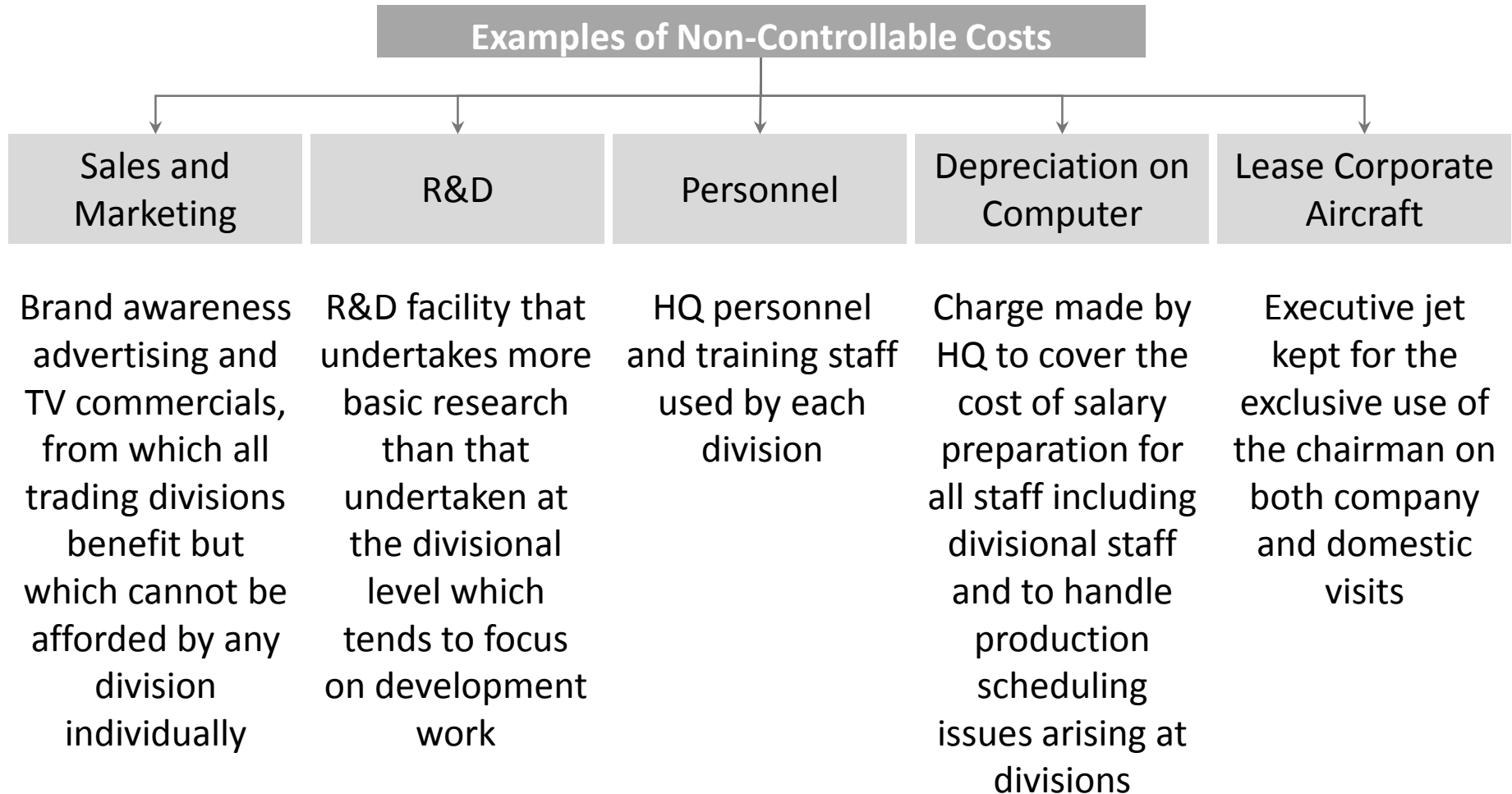


Consider the following divisional profit and loss account; note the share of costs incurred by the headquarters on behalf of all divisions

Division XYZ: Profit and loss account for year ended 30 September 20x4		
	€000s	€000s
Sales		9 000
Cost of sales		
Materials	2 000	
Labour	1 500	
Production overheads	500	
Depreciation on machinery	500	
Other costs		
Rent and rates	1 000	
Sales and marketing (divisional)	300	
1 Sales and marketing (share of HQ)	550	
R&D (divisional)	700	
2 R&D (share of HQ)	300	
Share of corporate HQ services		
3 Personnel	200	
4 Depreciation on computer	100	
5 Lease of corporate aircraft	100	
		7 750
Profit		1 250

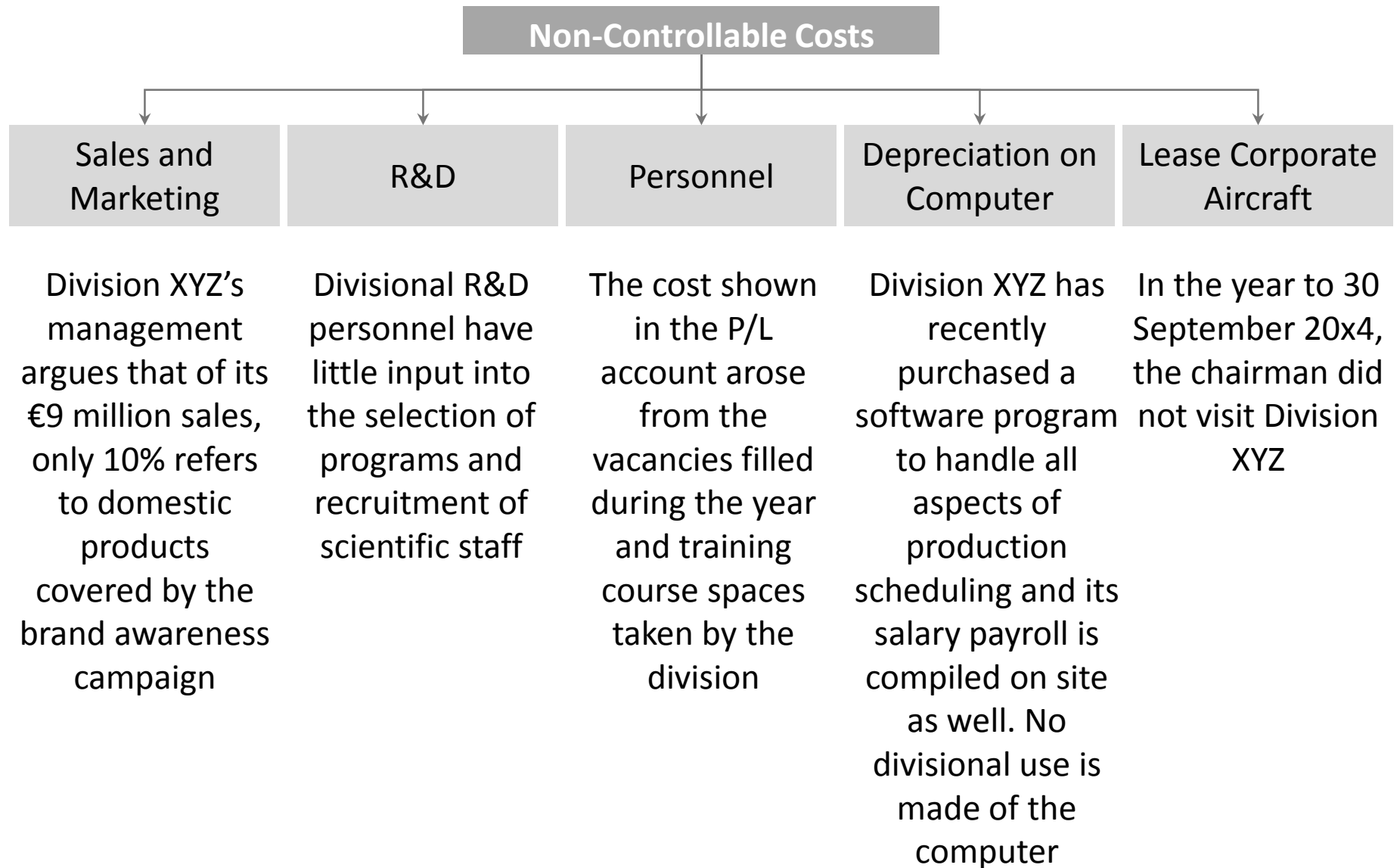
The traditional accounting profit of €1250 arises after deducting all costs (incurred and HQ-allocated) from revenue. But is it fair to hold the divisional manager responsible for all of these HQ-allocated charges?

This divisional share of HQ costs are considered by the divisional managers as non-controllable



It is advantageous to strip out from the divisional profit and loss account costs which were imposed on a division by corporate headquarters

Here, in our case we will consider each one of these costs...



...to come up with the following P/L account

Division XYZ: Profit and loss account for year ended 30 September 20x4

	€000s	€000s
Sales		9 000
Cost of sales		
Materials	2 000	
Labour	1 500	
Production overheads	500	
Depreciation on machinery	500	4 500
Gross profit		4 500
Other controllable costs:		
Rent and rates	1 000	
Sales and marketing (300 + 55)	355	
R&D	700	
HQ services: personnel	200	2 255
Controllable profit		2 245
Non-controllable costs		
Sales and marketing	495	
R&D	300	
Depreciation on computer	100	
Lease of corporate aircraft	100	995
Net profit		1 250

10% x €550

Division XYZ feels 'ownership' of this profit

Division's managers may argue that other costs deemed to be controllable but they are not controllable

Most assets situated physically within the division are expected to be controlled by divisional management except for:

1

Certain productive or research capacity made available to a division by either another division or by headquarters



example

An expensive piece of research equipment located in the central research laboratories is used exclusively on projects commissioned by a specific division



This should be allocated to the commissioning division's asset base

2

Divisions required to retain surplus assets for the future use of other parts of the company



example

Division XYZ is requested by HQ to hold on to a warehouse which had been freed up as a result of investment in just-in-time manufacturing procedures, this is in light of future plans elsewhere within the group



Value of the warehouse would have to be taken out of Division XYZ's asset base & allocated centrally

Agenda

14.1 Introduction

14.2 Why Divisionalize?

14.3 Types of Divisions

14.4 Defining Profits and Investments

14.5 Asset Base Valuation

14.6 Residual Income: An Alternative to ROI

14.7 The Imputed Rate of Interest Does Matter!

14.8 A Cautionary Note about Performance Measures

14.9 Transfer Pricing

14.10 Criteria for Establishing a Transfer Price

14.11 The International Dimension

Review Questions

Case Studies

For performance measurement purposes: assets are valued based on either their net book value or current replacement cost

Net Book Value

Difference between original purchase price of assets (both fixed & current) and accumulated depreciation

All other things being equal, a division's net book value of its asset base will fall year by year

A rising trend in **return on investment**

Current Replacement Cost

Using the most appropriate current value and not the depreciated historic cost

Through:

Suppliers' catalogues & price list

Industrial indices of specific price movements

The subjectivity involved in the process of determining replacement cost, together with the time taken on such an exercise, make companies use the net book value technique

But the net book value technique have a fatal shortcoming which is that it leads managers not to invest in additional assets to keep high ROI indicators

Example

A division reports annual controllable profits of €10 million. Its fixed asset base of €100 million remains the same in physical composition but is depreciated at 20% per annum straight-line. The following pattern emerges:

	Year 1	Year 2	Year 1
Divisional profit	€10m	€10m	€10m
Net book value of fixed assets	€100m	€80m	€60m
ROI	10%	12.5%	16.67%

Trend of increasing ROI

Inhibits divisional managers from:

Replacing ageing assets with modern equivalents

Investing in additional assets

Although

Such decisions is in the long-term interest of both the division & the company

Even if we factored in an increasing charge for maintaining ageing assets, thereby reducing marginally the divisional profit, the rising trend in ROI would still emerge

Agenda

14.1 Introduction

14.2 Why Divisionalize?

14.3 Types of Divisions

14.4 Defining Profits and Investments

14.5 Asset Base Valuation

14.6 Residual Income: An Alternative to ROI

14.7 The Imputed Rate of Interest Does Matter!

14.8 A Cautionary Note about Performance Measures

14.9 Transfer Pricing

14.10 Criteria for Establishing a Transfer Price

14.11 The International Dimension

Review Questions

Case Studies

Another example of ROI indicators leading to dysfunctional decisions

Consider the **dilemma** confronting the management of the division whose return on investment figures are displayed above

At the end of Year 3 management is presented with an investment opportunity which will, by itself, earn 12% ROI. Management is likely to reject such an opportunity because it will have the effect of watering down the already high ROI of 16.67%. But if the cost of capital within the group (that is, the average rate of interest paid by the group for borrowing funds externally including servicing the owner's equity by dividends) is only 10%

↓
The group would gain if this division accepted the investment proposal

ROI > Cost of Capital



In an attempt to overcome this problem of ROI, companies may use residual income (RI) measures

Divisions are charged interest on their invested capital



This interest is the rate of the company's cost of capital

Expressed in absolute money amounts



Management's goal should be to accept investment proposals which exceed the imputed interest charge

Example showing the difference between the ROI and the RI computations

Example

A company, comprising three divisions, charges each division 12% for the use of corporate resources for investment purposes. Each division's profit performance and investment base is given below.

	Division 1	Division 2	Division 3
Controllable profit	€100m	€240m	€95m
Investment in net assets	€350m	€850m	€300m
Performance measures			
ROI	28.57%	28.23%	31.66%
RI			
Controllable profit	€100m	€240m	€95m
Imputed interest:			
12% on net assets	42m	102m	36m
RI	€58m	€138m	€59m

$(€100 \div €350) \times 100$ points to the ROI calculation for Division 1.
 $€350 \times 12\%$ points to the imputed interest calculation for Division 1.

In terms of ROI, Division 3 is making the best use of its assets but with the imputed interest charge of 12 per cent, Division 2 is performing best of all

RI overcomes the dysfunctional element of ROI by encouraging divisional managers to invest, provided the expected returns exceed the imputed cost of capital...

Example (Continued)

Imagine that the management of Division 3 is presented with a proposal which will boost controllable profit by €15m for a local investment of €50m, an ROI of 30% which makes the proposal extremely attractive to corporate management whose cost of capital is 12%.

But the managers in Division 3 would oppose such a suggestion because they would witness a drop in their ROI:

$$\text{Before investment ROI: } \frac{€95\text{m}}{€300\text{m}} = 31.6\%$$

$$\text{After investment ROI: } \frac{€95\text{m} + €15\text{m}}{€300\text{m} + €50\text{m}} = 31.4\%$$

Although this drop is only marginal, Division 3 would wish to avoid the incremental investment; Division 3's ROI is ahead of the field and that is where its managers would want it to stay, particularly as managerial remuneration or promotion is based on ROI

...where RI allows local management to accept the proposal for the benefit of both the division and the company as a whole

RI Computation

$$\begin{aligned}\text{Before investment RI} &= \text{€95m less 12\% of €300m} \\ &= \text{€95m} - \text{€36m} \\ &= \text{€59m}\end{aligned}$$

$$\begin{aligned}\text{After investment RI} &= \text{€110m less 12\% of €350m} \\ &= \text{€110m} - \text{€42m} \\ &= \text{€68m}\end{aligned}$$

But still the RI faces two shortcomings

Two problems must be overcome before a company can use RI

- 1 It is not easy to calculate the cost of capital
- 2 The valuation of the asset base is once again a significant factor in the calculation

Agenda

14.1 Introduction

14.2 Why Divisionalize?

14.3 Types of Divisions

14.4 Defining Profits and Investments

14.5 Asset Base Valuation

14.6 Residual Income: An Alternative to ROI

14.7 The Imputed Rate of Interest Does Matter!

14.8 A Cautionary Note about Performance Measures

14.9 Transfer Pricing

14.10 Criteria for Establishing a Transfer Price

14.11 The International Dimension

Review Questions

Case Studies

This example shows that the selection of the interest rate does matter; the managers of these three divisions would have a vested interest in the rate selected for RI purposes

Example

A company has three divisions with different net assets bases and profit performances. Corporate HQ is undecided about the rate of interest to impute for the use of corporate resources. Depending on various factors, it has been advised that 8%, 12% or 15% could be adopted.

	Division Small	Division Medium	Division Large	
Controllable profits	€25m	€80m	€120m	
Net assets investment	€100m	€400m	€800m	
ROI	25%	20%	15%	
Ranking	(1)	(2)	(3)	
	€m	€m	€m	
Profits	25	80	120	
Imputed interest @ 8%	-8	-32	-64	
RI	17	48	56	Low interest rates favor divisions with high investment in net assets
Ranking	(3)	(2)	(1)	
Imputed interest @ 12%	-12	-48	-96	
RI	13	32	24	
Ranking	(3)	(1)	(2)	
Imputed interest @ 15%	-15	-60	-120	
RI	10	20	NIL	High interest rates do not favor divisions with high investment in net assets
Ranking	(2)	(1)	(3)	

Do-it-Yourself Case Study

Case Study

Compute plc is split into four operating divisions, Chip, PC, Printer and Drive. Each division is seen as being an autonomous unit with profit and investment responsibilities. The cost of capital for the company is 13%. In the ROI calculations for performance measurement purposes, the beginning-of-year asset base is used for each division. Fixed assets are depreciated on a straight-line basis.

The following positions are recorded as at 1 April 20x5.

	Investment in net assets 1 April 20x5 (€m)	Budget for year to 31 March 20x6 Controllable profit (€m)
Chip	192.0	48.0
PC	270.0	90.0
Printer	168.0	50.4
Drive	120.0	15.6

Do-it-Yourself Case Study (Continued)

After the divisional budgets for 20x5/x6 were drawn up, each division had additional investment proposals to consider which, if accepted, would alter the above figures because each would be completed by 1 April 20x5. The divisional proposals are as follows:

Chip

An investment of €60 million in new clean-room facilities for wafer fabrication which would produce sales of €90 million per annum. Expected net profit is €12 million per annum.

PC

Termination of the assembly of personal computers which is budgeted to yield a profit of €9 million for 20x5/x6. Anticipated revenue from sale of dedicated assets €45 million whose original cost seven years ago was €360 million. When purchased the division believed the equipment would last eight years.

Printer

Sale of entire production line of dot matrix printers for book value being €12 million. The budgeted figures above include a profit of €12 million from this line. Additionally, €60 million would be invested in a laser printer production line which would give €18 million profit per annum.

Drive

Investment in an additional production line to cope with increased demand for small-diameter disks; €48 million required, with expected sales of €21.6 million and annual profits of €6.72 million.

Do-it-Yourself Case Study (Continued)

As the financial consultant to Compute plc you are required to do the following:

1. On the basis that all proposals are implemented on 1 April 20x5:
 - (a) calculate the updated ROI for each division for the year to 31 March 20x6;
 - (b) identify those divisional managers who would expect to receive a higher bonus if bonuses are related to ROI;
 - (c) explain how the decisions of each divisional management may affect the company's overall financial performance for 20x5/x6.
2. Imagine that the company has an investment limit of €120 million for 20x5/x6. Which proposals would be selected?
3. Calculate the lowest price at which the personal computer assembly line should be sold by PC Division so that the company makes neither a profit nor a loss on the deal.
4. Calculate the budgeted RI for each division on the basis that each investment/disinvestment goes ahead.

Do-it-Yourself Case Study (Solution)

Requirement 1

	Chip	PC	Printer	Drive
Controllable profit per original budget 20x5/x6	€48.0m	€90.0m	€50.4m	€15.60m
Adjustments to profit	+ 12.0m	– 9.0m	+ 6.0m*	+ 6.72m
Revised profit	€60.0m	€81.0m	€56.4m	€22.32m
* + €18.0m – €12.0m = + €6.0m				
Net asset investment base per original budget 20x5/x6	€192.0m	€270.0m	€168.0m	€120.0m
Adjustment to base	+ 60.0m	– 45.0m	+ 48.0m**	+ 48.0m
	€252.0m	€225.0m	€216.0m	€168.0m
** + €60.0 – €12.0				
ROI per original budget	25%	33.33%	30%	13%
ROI as adjusted	23.81%	36%	26.11%	13.28%

On a straight comparison of changes in ROI the managers of Divisions PC and Drive will anticipate receiving an increased bonus as a result of the new proposals being implemented

Do-it-Yourself Case Study (Solution)

Requirement 1 (Continued)

An individual breakdown of investment returns can be compared with the company-wide cost of capital of 13%:

	Chip	PC	Printer	Drive
New profits	€12.0m	(€9.0)	€6.0m	€6.72
New investment	€60.0m	(€45.0)	€48.0m	€48.00
ROI	20%	(20%)	12.5%	14%

The opportunities presenting themselves to the divisional management of Chip & Drive, because they exceed the company's cost of capital, should be accepted; and not Printer & PC

But note that, if the investment decisions are left entirely to the discretion of local management, the management of PC and Drive & not Chip and Drive would accept it

Reason

Because their ROI performance measures will improve

	Chip	PC	Printer	Drive
ROI per original budget	25%	33.33%	30%	13%
ROI as adjusted	23.81%	36%	26.11%	13.28%

Decisions	Chip	PC	Printer	Drive
Divisional view	No go	Go	No go	Go
Company view	Go	No go	No go	Go

Because of that dysfunctional behavior emerging from the use of ROI, the management of Chip and PC would take decisions which harmed the financial position of the company

Do-it-Yourself Case Study (Solution)

Requirement 2

Management must select those projects which it feels fit with the overall strategic plan of the group. There are many factors to be taken into consideration in this selection process, including the financial implications, the following numbers would be considered:

Overall sum required	€156 million
Limit on funds	€120 million
Excess requirement	€ 36 million

Division	Funds required	ROI	RI
Chip	€60m	20%	€12m less $(13\% \times €60m) = €4.2m$
Printer	€48m	12.5%	€6m less $(13\% \times €48m) = (€0.24m)$
Drive	€48m	14%	€6.72m less $(13\% \times €48m) = €0.48m$

By adopting either ROI or RI management would select the proposals emanating from divisions Chip and Drive

Do-it-Yourself Case Study (Solution)

Requirement 3

The suggestion from PC Division that it sell off its assembly line was rejected:

Because it was currently earning 20% per annum against a cost of capital of 13%

Unless there was a strategic decision lying behind this proposal (for example, to source the assembly of PCs in the Far East) this would be a move which would penalize the group. The question asks how much would have to be offered to Compute plc before such a deal would make some sense. That is:

What price would have to be received for the assembly line before the company made neither a profit nor a loss?

The proposal would therefore require the sale proceeds of plant to drop to a point where the loss to the company was equivalent to 13% ROI:



$$\begin{array}{lcl} \text{€9m lost proceeds} & = & 13\% \text{ of } X \\ X & & - \text{€69.23 million} \end{array}$$

Do-it-Yourself Case Study (Solution)

Requirement 4

	Chip £m	PC £m	Printer £m	Drive £m
Revised profit forecast	60.00	81.00	56.40	22.32
Imputed interest @ 13%	32.76	29.25	28.08	21.84
RI	<u>27.24</u>	<u>51.75</u>	<u>28.32</u>	<u>0.48</u>

Agenda

14.1 Introduction

14.2 Why Divisionalize?

14.3 Types of Divisions

14.4 Defining Profits and Investments

14.5 Asset Base Valuation

14.6 Residual Income: An Alternative to ROI

14.7 The Imputed Rate of Interest Does Matter!

14.8 A Cautionary Note about Performance Measures

14.9 Transfer Pricing

14.10 Criteria for Establishing a Transfer Price

14.11 The International Dimension

Review Questions

Case Studies

If top management make economic and personnel decisions based solely on ROI or RI it will be deluding itself about the underlying talents of its divisional teams

Example

- A division in an integrated group of companies may consistently report losses, a negative ROI and RI, and therefore a decision to replace management may appear to be called for.
- But behind these indicators may lie a superior managerial performance which has reduced losses in a strategically crucial activity

Many other indicators of performance must be considered alongside the financial ones



Peters and Waterman (1982)

Study: In Search of Excellence: Lessons from America's Best Run Companies



Well-managed companies have their corporate eyes on managerial performance indicators as well as the financial indicators

Agenda

14.1 Introduction

14.2 Why Divisionalize?

14.3 Types of Divisions

14.4 Defining Profits and Investments

14.5 Asset Base Valuation

14.6 Residual Income: An Alternative to ROI

14.7 The Imputed Rate of Interest Does Matter!

14.8 A Cautionary Note about Performance Measures

14.9 Transfer Pricing

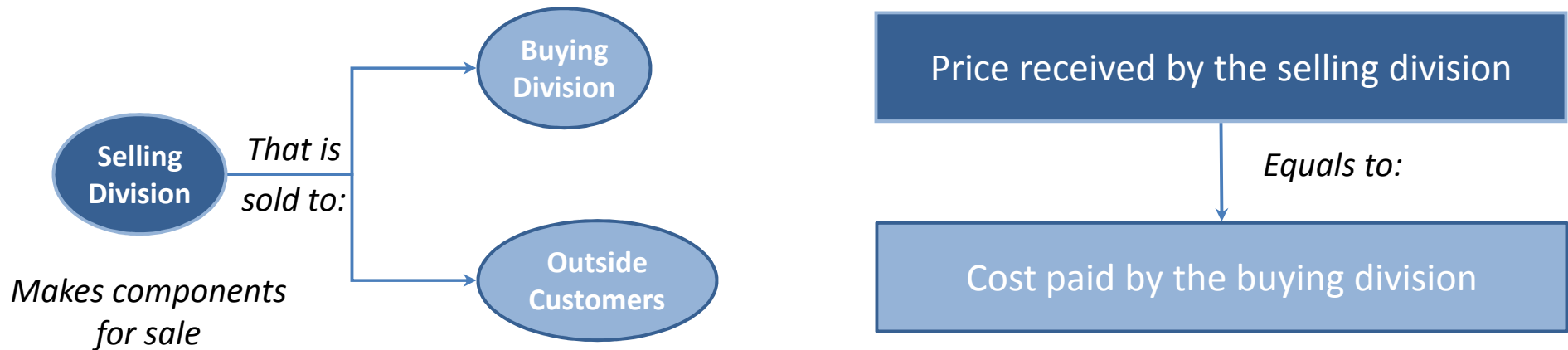
14.10 Criteria for Establishing a Transfer Price

14.11 The International Dimension

Review Questions

Case Studies

We'll now consider trading between divisions within the one company and how the price is established for such a transfer of goods and services



If the selling division sets the price to give itself a handsome mark-up on cost, this price will negatively impact the purchasing division

Then:

The ROI calculations one will look superior to the other even though an external customer has not been involved

Agenda

14.1 Introduction

14.2 Why Divisionalize?

14.3 Types of Divisions

14.4 Defining Profits and Investments

14.5 Asset Base Valuation

14.6 Residual Income: An Alternative to ROI

14.7 The Imputed Rate of Interest Does Matter!

14.8 A Cautionary Note about Performance Measures

14.9 Transfer Pricing

14.10 Criteria for Establishing a Transfer Price

14.11 The International Dimension

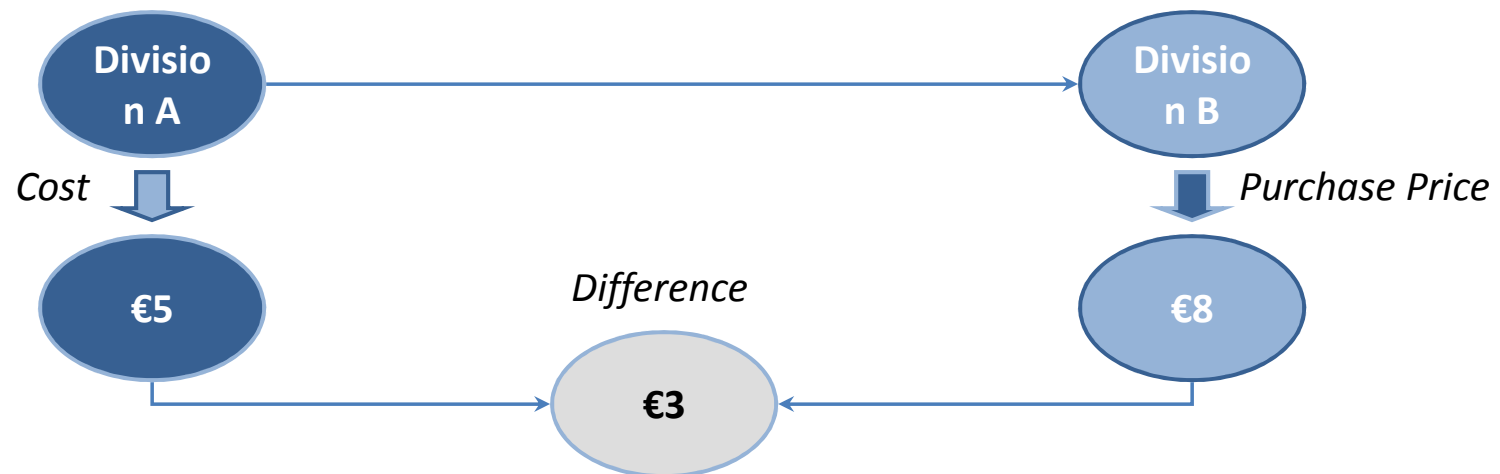
Review Questions

Case Studies

Divisional managers are free to do business with each other provided the interests of the divisions are congruent with those of the company

Example

- Division A makes and sells a component to external customers for €10 each.
- Division B has a need for this component for its own manufacturing process which it could buy for €8 from outside the group.
- Division A refuses to drop its price for an internal transfer and Division B buys outside.
- Such an outcome follows the principle of giving divisional managers complete autonomy in the running of their affairs. Spare capacity exists in Division A.
- But an examination of the costs of Division A's component reveals that only €5 were spent on making it and Division A was making a profit of a further €5 per unit sold.
- Therefore from the company's point of view the company is €3 worse off on the decision of the two divisions not to do business with each other.



Two prices are possible for transferring goods and services between the two divisions: market prices and cost-based prices

Market Prices

The best method for transferring goods and services between divisions

It can be objectively tested by both parties to the deal

To determine the price, we use:

Suppliers' price lists, quotations or market observations

Exceptions:

- When market price is dumped below costs
- When quality required exceeds that of the market; thus requiring a premium

€8

Cost-Based Prices

The second best: resort when comparable market prices are not easily determined

To determine the price, we use:

Three types of costs:

1. Full Costs

Variable costs plus a proportion of fixed costs (normal absorption formula of the division)

2. Variable Costs

Overcomes full Cost's shortcomings

3. Negotiated Costs

When market price and cost-based prices have not worked

Full cost usually seem the most appropriate cost-based price, as it would appear to prevent the selling division from losing money on the deal, but still some concerns rise

Shortcomings of Full Costing

It gives the selling division more control over the price by basing its absorption of fixed costs on a low denominator volume of activity (leading to a high fixed cost per unit)



It is appropriate to allow the buying division's accountants to audit the selling division's cost structures prior to agreement

Example (Continued)

Division A	€
Variable cost	3
Fixed cost	2
Full cost	<u>5</u>
Profit	5
Selling price	<u><u>10</u></u>

Division B
Variable cost
Division A's component
Material and labour
Fixed cost
Full costs

€
5
10
5
<u>20</u>

If B's management resorts to pricing on variable cost only; the lowest price it could charge would be €15

But in terms of company variable cost, it could afford to drop to €13 (Division A, €3, plus Division B, €10); this wrong signals (between €15 & €13) could lead to loss of sales

As just shown, variable costing would overcome the shortcoming of full costing especially if there's a distress in the market

Shortcomings of Variable Costing

Its use leaves the selling division (Division A) neither up nor down on the deal and it must sit back and watch Division B making all the profit on the ultimate sale



Why should Division A be penalized in this way, particularly if there is a buoyant market for its goods?

Negotiated costing can be considered like the last resort for the HQ to set the conflict between the divisions on the price

Two Prices are used

Division A selling to Division B at €10 and Division B buying from
Division A for €8



Creation of an HQ Contingency Account

An account held by the HQ that includes the difference between both prices

This dual pricing procedure cannot be adopted for a lengthy period of time otherwise the motivation of both divisions, and their confidence in the system, will deteriorate

Agenda

14.1 Introduction

14.2 Why Divisionalize?

14.3 Types of Divisions

14.4 Defining Profits and Investments

14.5 Asset Base Valuation

14.6 Residual Income: An Alternative to ROI

14.7 The Imputed Rate of Interest Does Matter!

14.8 A Cautionary Note about Performance Measures

14.9 Transfer Pricing

14.10 Criteria for Establishing a Transfer Price

14.11 The International Dimension

Review Questions

Case Studies

Where divisions in one company cross international boundaries, there are other factors to be considered; first issue is the taxation angle

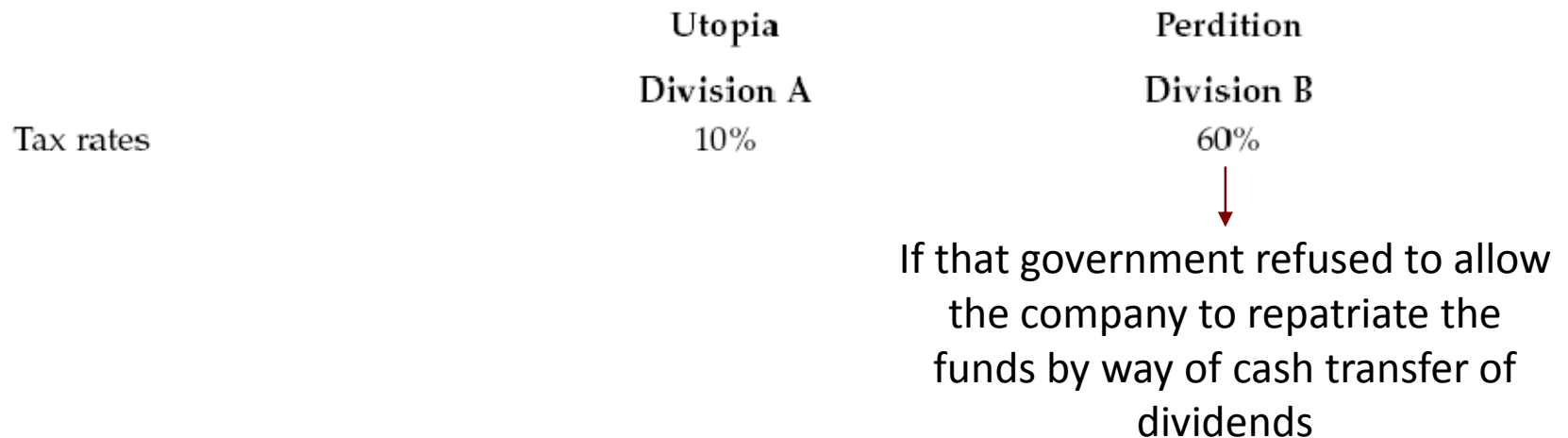
Example

Imagine that a company has two divisions in two separate countries, Utopia and Perdition, and that Division A passes components to Division B.

<i>Scenario 1</i>	Utopia	Perdition
	Division A	Division B
Tax rates	10%	60%
Transfer price selected	High	
Effect	High earnings from sales to Division B are subjected to low taxes	High costs of bought-in goods from Division A reduce taxable profits
<i>Scenario 2</i>		
Tax rates	60%	10%
Transfer price selected	Low	
Effect	Low earnings from sales to Division B are reported for tax purposes	Low bought-in costs produce high taxable profits (at a low rate of tax)

A company very often wants to minimize the amount of tax it pays on profits, this is to reduce cash outflows and boost earnings per share

Second is the issue of repatriation of profits



Here transfer prices could be adjusted so that the division located in the restricting country is asked to pay high transfer prices; this way the cash emerges from the country by way of transfer prices rather than dividends which could be blocked

Do-it-Yourself Case Study

- The Ten Ton Trailer Company comprises two divisions, one of which is the Freezer division which manufactures freezing equipment for refrigerated containers.
- Some of the freezer units are sold directly by the Freezer division to outside customers.
- The rest are passed to the Box division which incorporates the freezer units into each of its containers for sale to outside customers.
- Both divisions have been set up as profit centers and have had a long history of rivalry about which is the more profitable; the question of transfer prices charged by the Freezer division has always been a sensitive issue.
- For some time now the market price of freezing units has been used because a number of competitors are active in the UK and Continental Europe.

Do-it-Yourself Case Study (Continued)

The following information is available to both divisions' management.

	€
Selling price for completed container	33 750
Selling price for freezer unit from Freezer division	22 500
Freezer division's costs per unit	13 500
Box division's costs per unit	16 875

For his part, the manager of the Box division is concerned with the way his numbers look.

	€	€
Selling price		33 750
Cost: Bought-in component	22 500	
Own division	16 875	39 375
Loss per container		5 625

Do-it-Yourself Case Study (Continued)

You are required to address the following issues:

1. If there is no surplus capacity should the Freezer division be required to transfer freezer units to the Box division? If so, is the market price (as currently used) the most appropriate one?
2. Present space limitations restrict Freezer division from making more than 1400 units per month, of which 1100 units are sold to external customers. Should the balance be transferred to Box division, and if so, at what price?
3. On the basis that Freezer division decided to charge a transfer price of €16,875 for up to 300 units, calculate the contribution to The Ten Ton Trailer Company. Would you, as manager of Box division, buy at €16,875?

Do-it-Yourself Case Study (Solution)

Requirement 1

Freezer division should certainly not be required to transfer units to Box. Freezer division is making more money on units sold to external customers than the company is making on the container.

	Company Position		
	Freezer unit	Container	
Sales price	€22 500	€33 750	
Costs	13 500	30 375	(13 500 + 16 875)
Profit	<u>€ 9 000</u>	<u>€ 3 375</u>	

The market price-based transfer price would appear to be the best available transfer price because €22,500 is the cash that would be foregone if Freezer division redirected a unit away from the market towards Box division

Do-it-Yourself Case Study (Solution)

Requirement 2

- As soon as excess capacity can be detected in Freezer division the €22,500 market price becomes invalid for use as a transfer price for those units that cannot be disposed of externally
- The divisional costs of €13,500 would seem to be the best price to settle at in the absence of additional information.
- In this way Freezer division breaks even on the surplus 300 units and Container division makes a profit of €3,375 as outlined above on each of the 300 surplus units.

Do-it-Yourself Case Study (Solution)

Requirement 3

If Box division accepted this transfer price the following picture emerges for each of the surplus 300 units.

	Freezer unit	Container	
Sales price	€16 875	€33 750	
Costs	13 500	33 750	(16 875 + 16 875)
Profit	€ 3 375	€ 0	

Container division would 'take the hit' on profits but the company as a whole would earn a profit of €3,375. Indeed, if the transfer price for the surplus 300 units lay between €13,500 and €16,875 each, this would lead to a beneficial outcome for the company.



But should Freezer division be allowed to keep all of the €, profit or should it perhaps be distributed? Otherwise Box division has little incentive to buy at €16,875

Case Study 14.1

Spectrum Technology PLC manufactures sophisticated electronic test equipment for several markets. The company is organized in three independent operating divisions to tackle these markets – Medical Products, Defense Products and Scientific Products.

For the year just ended, the divisional financial performance was as follows:

	Medical	Defence	Scientific
	€000	€000	€000
Sales	1 850	1 240	780
Variable costs	1 070	560	450
Contribution	<u>780</u>	<u>680</u>	<u>330</u>
Fixed costs	410	370	265
Net profit	<u>370</u>	<u>310</u>	<u>65</u>
Net assets	<u>1 200</u>	<u>800</u>	<u>550</u>

Case Study 14.1 (Continued)

- Spectrum Technology The central R&D laboratory has recently completed the development of a new product which is targeted on a new segment in monitoring quality control in the food processing sector. At this stage, the company is reluctant to set up a new division to handle this new market area and its directors have decided that one of the divisions must take on the responsibility for this product line.
- The new product will require an investment in new plant and extra working capital of €0.3m. However, market research has indicated that first year sales should reach €0.25m, producing an increase in contribution of €0.17m, but requiring extra fixed costs of €80,000 per annum.

Case Study 14.1 (Continued)

- In reviewing the purchasing activities of the divisions, the Group Managing Director has noted that the Defense Division (DD) is currently purchasing a specialized, single sourced, optical component X–Y2A from an outside supplier for €2,250 per unit. DD is unaware that the Medical Division (MD) also has a considerable internal requirement for this component and has commenced to manufacture it in quantity with existing facilities in order to avoid any over-dependency on the single supplier. The Group Managing Director is keen that DD transfer its procurement to within the group, but the MD Divisional Manager has indicated that his costings show that the price to DD should be €2,552 per unit.

Component X–Y2A – Costing	€
Material	1 110
Labour	410
Variable overheads	600
Fixed overheads	200
Full cost	<u>2 320</u>
Profit margin 10%	<u>232</u>
Selling price	<u>2 552</u>

Case Study 14.1 (Continued)

Required

1. If the performance of the managers of each division is being assessed by the ROI method, which of the divisions would wish to incorporate the new product line into its divisional activities?
2. If the company measures divisional performance in terms of the RI method, would the conclusion alter on the acceptability of the new product line? The company has assessed risk level in each of the divisions and decided upon capital costs being charged to the Medical Division at 16 per cent, to the Defense Division at 20% and to the Scientific Division at 12%.
3. If the current market price for the X–Y2A component remains unaltered, the Managing Director will have difficulty in effecting its internal procurement by the Defense Division from the Medical Division. If the group is attempting to formulate a transfer pricing policy, which of the potential bases available would be consistent with this internal procurement going ahead?

Case Study 14.1 (Solution)

Requirement 1

To assess the acceptability of the new product line to existing divisions, it is necessary to compare existing ROI per division with the projected ROI from the new product line.

ROI computation:

$$\text{Medical Division: } \frac{€370}{€1\,200} = 30.83\%$$

$$\text{Defence Division: } \frac{€310}{€800} = 38.75\%$$

$$\text{Scientific Division: } \frac{€65}{€550} = 11.82\%$$

$$\text{New product (Food processing): } \frac{€90}{€300} = 30.0\%$$

Neither the Medical Division nor the Defense Division will take on the new product line since its inclusion in their results would have the effect of diluting their performance; unlike the Scientific Division as it generates an increase of 30% ROI

Case Study 14.1 (Solution)

Requirement 2

If divisional performance is monitored by RI, then it is necessary to compute existing RI for each division, taking into account the different costs of capital involved.

Current RI computations:

	€000
Medical Division	
Profit	370
Less: Imputed interest (16% of €1200)	192
Residual income	<u>178</u>
Defence Division	
Profit	310
Less: Imputed interest (20% of €800)	160
Residual income	<u>150</u>
Scientific Division	
Profit	65
Less: Imputed interest (12% of €550)	66
Residual income	<u>(1)</u>

Case Study 14.1 (Solution)

RI of new product

Additional contribution less extra fixed costs = €170 000 – €80 000 = €90 000

But, imputed interest varies between the different divisions.

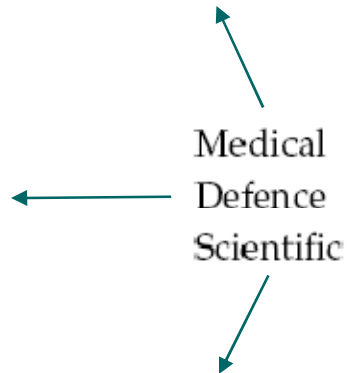
	€000
Medical Division	
Additional net contribution	90
Less: Imputed interest (16% of €300)	48
Additional residual income	<u>42</u>
Defence Division	
Additional net contribution	90
Less: Imputed interest (20% of €300)	60
Additional residual income	<u>30</u>
Scientific Division	
Additional net contribution	90
Less: Imputed interest (12% of €300)	36
Additional residual income	<u>54</u>

Case Study 14.1 (Solution)

In each division, RI will be enhanced by the introduction of the new product line – summarized below:

They might wish to take on the new product line to consolidate their position as the leading division

They would see an opportunity to catch up and just exceed the Medical Division as the premier division, with a revised RI of €180,000 against the Medical Division's unchanged RI of €178,000



They would be desperate to have the new product line to move into a position of positive RI

Current RI €000	New product RI €000	Revised RI €000
178	42	220
150	30	180
(1)	54	53

If there were a switch from ROI to RI divisional assessment, the divisions' response to this investment opportunity would be strikingly different

Case Study 14.1 (Solution)

Requirement 3

- The emerging situation with component X-Y2A highlights a classic example of conflict of interest between divisions, with the interests of the company as a whole likely to suffer at the end.
- MD is only interested in internal manufacture and hence its divisional manager is happy to balance the extra unit cost against security of supply. Even so, MD wishes to make a margin to enhance either its ROI or RI assessment if sales are made to DD. However, DD will choose to continue to buy from the outside supplier since €2,250 is less than the price MD would wish to charge. Again, DD is concerned to maximize ROI or RI.
- The fact that the Group Managing Director has become involved in the issue will, almost inevitably, mean that an 'imposed' solution will result. For DD to accept supplies from MD, it will mean that the component will have to be transferred at €2,250 or below. If the group wishes to install a definite transfer pricing policy, then the only policy consistent with DD accepting this supply is a variable cost only policy or €2,120 per component. However, the Group Managing Director will have to expect demotivational consequences for MD, as a trade-off against the benefit of reduced costs for the company.